

Integrating Driving-Aware World Model With MPC for Autonomous Driving at Unsignalized T-Intersections

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Abstract—Model Predictive Control (MPC) is widely used in autonomous driving for its robustness and control effectiveness. However, its reliance on explicit physical models can lead to mismatches in interactive scenarios, resulting in trajectory deviations and safety risks. While recent world model approaches alleviate this limitation by learning latent dynamics from data, their learned representations often retain task-irrelevant background features, leading to instability propagation in downstream control policies. To address these challenges, we propose an end-to-end framework that integrates a Driving-Aware World Model (DAWM) with Model Predictive Path Integral (MPPI) control. DAWM employs a spatial prior and a scoring network to adaptively reweight Vision Transformer (ViT) patch embeddings, producing compact and task-relevant latent states. It further introduces an incremental latent dynamics model to predict state transitions, and an uncertainty-aware loss weighting mechanism to balance multi-objective training. Meanwhile, MPPI sampling is guided by a reinforcement learning policy and augmented with artificial potential fields (APFs) to enhance safety and convergence. In unsignalized T-intersection scenarios in CARLA, our approach outperforms standard MPPI by 66.7% in task success rate, while also reducing collisions and lane deviation, highlighting its superior control performance.

Index Terms—Autonomous driving, world model, MPPI, vision transformer, reinforcement learning.

I. INTRODUCTION

AUTONOMOUS driving, as a critical component of intelligent transportation systems, aims to achieve vehicle automation through the effective combination of perception, decision-making, and control [1], [2]. However, the deployment of autonomous driving systems in complex scenarios remains a significant challenge [3]. One representative example is driving in unsignalized intersections, which are characterized by high dynamism, uncertainty among multiple participants, and potential conflict risks [4], [5], [6]. These factors pose substantial demands on the perception and planning capabilities of autonomous systems. For instance, in Fig.1, vehicles must operate without explicit traffic signals or

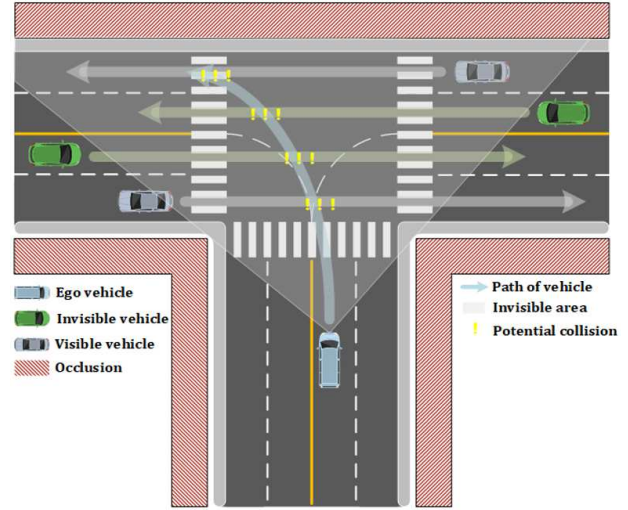


Fig. 1. An illustration of a typical unsignalized T-intersection.

instructions, relying solely on environmental information to make real-time decisions [7], [8], [9]. The primary goal is to balance safety and efficiency, which requires addressing the intricate dynamics of interacting agents while mitigating risks in uncertain and rapidly changing conditions [10].

To address the control challenges in complex scenarios, Model Predictive Control (MPC), a receding-horizon optimization strategy with a finite time horizon, has been extensively applied in autonomous driving [11], [12]. Model Predictive Path Integral (MPPI) can be seen as a sampling-based version of MPC [13]. Unlike traditional MPC, MPPI performs trajectory rollouts and iterative optimization in a virtual manner through model-based propagation, without directly solving the optimization problem in closed form. However, the challenge of learning accurate dynamics models and the computational burden of high-dimensional inputs together constrain MPPI performance [14], [15].

Recent works have explored world models that capture the environment's dynamics to support trajectory prediction and control optimization [16], [17], [18]. These world models typically consist of two key components: a representation model that encodes high-dimensional observations into compact latent states, and a latent dynamics model that predicts the temporal evolution of these latent states over time steps [19]. Building upon this framework, [20] proposes the Task-Oriented Latent Dynamics (TOLD) model,

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an implicit (decoder-free) world model trained end-to-end to capture features that are most relevant to the control objective. Despite recent advancements, existing world models still face limitations in complex driving scenarios due to architectural and training constraints. In terms of representation, most world models employ convolutional architectures. While effective at capturing local spatial features, they are limited in modeling long-range dependencies—such as the relative positions of distant vehicles and the overall road layout [21]. In contrast, Vision Transformer (ViT)-based models leverage self-attention mechanisms to capture long-range dependencies. However, their attention mechanism lacks strong inherent spatial inductive bias, which may limit the model's ability to focus on task-relevant regions in driving scenarios [22]. In terms of prediction, latent dynamics models aim to predict sequential future states from compact latent representations. Directly predicting full next states at each step not only increases the learning difficulty—requiring accurate mappings under complex and uncertain driving scenarios—but also leads to error accumulation over long horizons. Such accumulated errors may significantly undermine the reliability of long-horizon predictions. In terms of optimization, balancing multi-task objectives in joint world model training remains a key challenge. Fixed or manually tuned loss weights require extensive hyperparameter search and cannot adapt to changing task difficulties or training dynamics, which may hinder convergence and generalization [23].

To address the challenges mentioned above, we propose a novel end-to-end framework for autonomous driving in complex scenarios. The framework leverages an improved world model to encode front-view images into task-relevant latent states and model their temporal evolution for long-horizon trajectory prediction in latent space. A sampling-based MPPI controller is employed, whose sampling process is guided by a reinforcement learning (RL) policy to accelerate convergence and augmented with artificial potential fields (APFs) to enhance both safety and robustness. Our contributions are summarized as follows:

- 1) We introduce an end-to-end driving framework integrating the Driving-Aware World Model (DAWM) with RL-guided MPPI sampling, enabling efficient and robust control in complex driving scenarios.
- 2) We incorporate a spatially-adaptive token reweighting scheme into the representation model, where multi-scale spatial priors guide the ViT architecture to focus more effectively on task-relevant regions.
- 3) To improve prediction stability and accuracy, we adopt an incremental latent dynamics model that predicts latent state increments rather than full next states.
- 4) We propose a principled uncertainty-aware loss weighting mechanism that dynamically balances multi-task objectives, thereby facilitating faster convergence and improved overall performance.

II. PRELIMINARIES

A. Problem Statement

Consider a general constant discrete dynamical system

$$x_{t+1} = f(x_t, v_t) \quad (1)$$

where $x_t \in \mathbb{R}^n$ is the state of the system at time t , and $v_t \in \mathbb{R}^m$ is the vector-valued control input at time t . In our driving task, v_t comprises both the vehicle's acceleration and steering angle. $f(\cdot)$ denotes the system dynamics, which can be a general nonlinear and input-nonaffine system. We assume that the control input v_t obeys a diagonal Gaussian distribution:

$$v_t \sim \mathcal{N}(u_t, \Sigma), \quad (2)$$

where $u_t \in \mathbb{R}^m$ is the commanded input. Σ is the covariance matrix. Without loss of generality, the covariance matrix can be infinitesimal to cover the case where the optimal policy is deterministic.

Assuming a finite time range $t \in \{t_0, t_0 + 1, \dots, t_0 + T - 1\}$, with each time unit determined by the system's control frequency. Sampling v_t at each time point, we can obtain a control sequence:

$$(v_{t_0}, v_{t_0+1}, \dots, v_{t_0+T-1}) = V \in \mathbb{R}^{m \times T}$$

For any initial state x_{t_0} , a unique state trajectory $X_V = \{x_{t_0+1}, x_{t_0+2}, \dots, x_{t_0+T}\}$ can be generated by executing V in (1). Assuming that $\mathbb{P}$ and $\mathbb{Q}$ are uncontrolled and controlled measures respectively, we can then define them by their density functions:

$$\begin{aligned} p(V) &= \prod_{t=0}^{T-1} \frac{1}{((2\pi)^m |\Sigma|)^{1/2}} \exp\left(-\frac{1}{2} v_t^T \Sigma^{-1} v_t\right) \\ q(V|U) &= \prod_{t=0}^{T-1} \frac{1}{((2\pi)^m |\Sigma|)^{1/2}} \\ &\quad \times \exp\left(-\frac{1}{2} (v_t - u_t)^T \Sigma^{-1} (v_t - u_t)\right). \end{aligned}$$

Using the running cost function $L(x_t, u_t)$ and a terminal cost $\phi(x_T)$, we can define the finite-horizon optimization problem as:

$$U^* = \arg \min_{U \in \mathcal{U}} \mathbb{E}_{Q_{U, \Sigma}} \left[\phi(x_T) + \sum_{t=0}^{T-1} \mathcal{L}(x_t, u_t) \right]. \quad (3)$$

where $\mathcal{U}$ is the set of allowed control sequences. We assume that the running cost $\mathcal{L}$ can be divided into an state-dependent running cost and a control cost:

$$\mathcal{L}(x_t, u_t) = c(x_t) + \frac{\lambda}{2} (u_t^T \Sigma^{-1} u_t + \beta_t^T u_t). \quad (4)$$

$S(V)$ is used to denote the portion of the trajectory cost that depends only on the state:

$$S(V) = \phi(x_T) + \sum_{t=0}^{T-1} c(x_t). \quad (5)$$

B. Model Predictive Path Integral

In this section, we provide a brief introduction to the key concepts of MPPI from an information theory perspective. More details can be found in previous research [24]. First, we can define the free-energy of the control system as:

$$\mathcal{F}(S, p, \mathbf{x}_0, \lambda) = -\lambda \log \left(\mathbb{E}_P \left[\exp \left(-\frac{1}{\lambda} S(V) \right) \right] \right). \quad (6)$$

where $\lambda \in \mathbb{R}^+$ is called the inverse temperature. By applying Jensen's inequality and the definition of KL-divergence, the above equation can be simplified as:

$$\begin{aligned} \mathcal{F}(S, p, \mathbf{x}_0, \lambda) &\leq \mathbb{E}_{Q_{U,\Sigma}}[S(V)] + \lambda \mathbb{D}_{\text{KL}}(Q_{U,\Sigma} \parallel \mathbb{P}) \\ &= \mathbb{E}_{Q_{U,\Sigma}} \left[\phi(x_T) + \sum_{t=0}^{T-1} \mathcal{L}(x_t, u_t) \right]. \end{aligned} \quad (7)$$

which denotes the free-energy as a lower bound on the state cost under the control distribution plus the KL-divergence between the base and controlled distribution. Hence, determining a control distribution that achieves this lower bound minimizes the expected cost and control cost [25].

As proposed in [24], we can now derive the optimal control input by minimizing the KL divergence between Q and Q^* :

$$\mathbb{E}_{Q^*}[v_t] = \int q^*(V) v_t dV. \quad (8)$$

While the optimal approach to resolving (8) involves direct sampling from Q^* , such an approach proves to be impractical. Importance sampling is therefore employed to compute the integral over the known distribution Q [26]:

$$\begin{aligned} \mathbb{E}_{Q^*}[v_t] &= \int \frac{q^*(V)}{q(V|U)} q(V|U) v_t dV \\ &= \mathbb{E}_Q[\omega(V) v_t], \end{aligned} \quad (9)$$

where the importance weighting term is:

$$\begin{aligned} \omega(V) &= \left(\frac{q^*(V)}{q(V|U)} \right) = \left(\frac{q^*(V)}{p(V)} \right) \left(\frac{p(V)}{q(V|U)} \right) \\ &= \frac{1}{\eta} \exp \left(-\frac{1}{\lambda} \left(S(V) + \frac{\lambda}{2} \sum_{t=0}^{T-1} u_t^T \Sigma^{-1} (u_t + 2\epsilon_t) \right) \right). \end{aligned} \quad (10)$$

where η is the normalizing constant. The control from (9) is globally optimal in an information-theoretic sense if the expectation is perfectly evaluated.

C. Temporal Difference Learning

RL aims to derive an optimal policy through interactions with an environment, typically formulated as a Markov Decision Process (MDP). An MDP is defined as $(\mathcal{S}, \mathcal{A}, T, R, \gamma)$, where $\mathcal{S}$ and $\mathcal{A}$ are the state and action spaces, T is the transition function, R is the reward function, and γ is a discount factor [27]. The objective is to find a policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ that maximizes the expected discounted cumulative rewards:

$$\mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(s_t)) \right]. \quad (11)$$

Temporal Difference (TD) learning algorithms aim to estimate the optimal state-action value function $Q^* : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, which quantifies the expected return when starting from a specific state-action pair and following the optimal policy thereafter. The function $Q^*(s, a)$ satisfies the Bellman optimality equation [28]:

$$Q^*(s, a) = \max_{a'} \mathbb{E}[R(s, a) + \gamma Q^*(s', a')]. \quad (12)$$

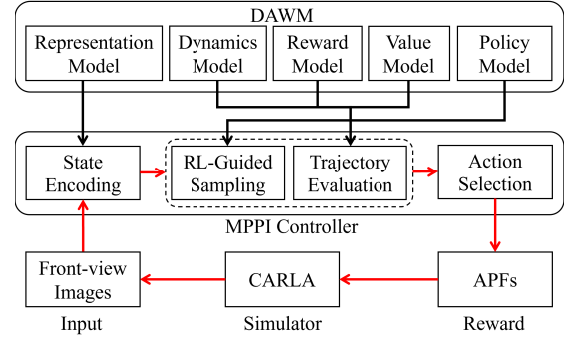


Fig. 2. Overview of the proposed framework. The red arrows denote the forward pipeline from perception to control, while the black arrows indicate the usage of different DAWM components within corresponding modules. The dashed box highlights the modules that are iteratively updated during each control cycle.

TD learning approximates Q^* using a parameterized function $Q_{\theta}(s, a)$ and updates θ iteratively by minimizing the temporal difference error:

$$\theta_{k+1} = \arg \min_{\theta} \mathbb{E}_{(s,a,s') \sim \mathcal{B}} [(Q_{\theta}(s, a) - y)^2], \quad (13)$$

where $y = R(s, a) + \gamma \max_{a'} Q_{\theta}(s', a')$. This method efficiently learns policies without requiring a transition model, making it well-suited for complex environments.

III. END-TO-END AUTONOMOUS DRIVING FRAMEWORK

In this section, we present the proposed end-to-end framework and outline the roles of its core modules, as illustrated in Fig. 2. Front-view RGB images are first processed by the representation model to produce a compact latent representation. The MPPI controller then iteratively refines a control distribution based on this representation. Guided by an RL policy, the controller samples multiple candidate control sequences from the initial distribution, and DAWM performs virtual forward rollouts over the planning horizon to predict future latent trajectories. APFs are embedded in the reward function to score each trajectory and drive subsequent distribution updates. After multiple iterations of optimization, the MPPI controller samples control sequences from the updated distribution and executes the first control command on the vehicle for interaction with the environment. In the following, we provide detailed descriptions of the design and implementation of each module.

A. MPPI Controller

The MPPI controller integrates the proposed Driving-Aware World Model (DAWM) to improve real-time adaptability and sample efficiency. As illustrated in Eq. (18), DAWM comprises five core components that together enable environment modeling and control guidance. The representation model encodes the current observation into a compact latent state, which serves as the input for downstream prediction. The latent dynamics, reward, and value models are used to model the environment in the latent space and enable trajectory rollouts. The policy model is employed to initialize the MPPI sampling distribution.

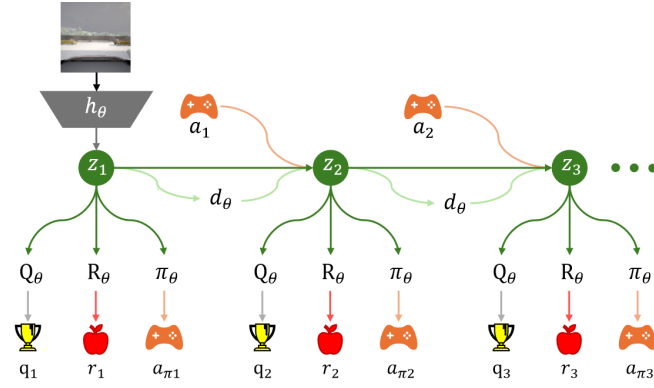


Fig. 3. An illustration of the process for inferring trajectory returns based on the current state z_1 , the world model, and a sampled sequence of actions $\{a_1, a_2, \dots\}$.

MPPI can be considered as a sampling-based MPC, where it utilizes importance sampling to iteratively update the action distribution. In practice, we fit parameters of a time-dependent multivariate Gaussian with diagonal covariance. Suppose we have an action sequence of length H , $\{a_t, a_{t+1}, \dots, a_{t+H-1}\}$. For the current state s_t , we need to use the world model to predict the corresponding reward $\{r_t, r_{t+1}, \dots, r_{t+H-1}\}$. First, we utilize the representation model to transform s_t into a latent state z_t . This step is designed to extract features and reduce the input dimensionality, thereby minimizing the number of model parameters. The latent dynamics model and reward model then use z_t and a_t to infer r_t and z_{t+1} , allowing the model to recursively infer a trajectory of length H . In addition, we utilize the replay buffer to further train a policy model π_θ . The policy function π_θ provides an action distribution P based on the current latent state z_t . By sampling actions from π_θ , we can leverage the latent dynamics model and reward model to subsequently infer the corresponding trajectories in a consistent manner. These trajectories help improve the convergence efficiency of the MPPI algorithm. The detailed process of trajectory rollout is shown in Fig.3.

Assuming the optimal action distribution P^* ,

$$\begin{aligned} u_i^* &= \mathbb{E}_{P_{U^*, \Sigma^*}}(v_i), \\ \text{diag}[\sigma_i^*] &= \sqrt{\mathbb{E}_{P_{U^*, \Sigma^*}}[(v_i - u_i) \circ (v_i - u_i)]}, \\ i &= t, \dots, t + N - 1, \end{aligned} \quad (14)$$

as noted in Section II, directly sampling from P^* is not feasible. The solution provided by MPPI is to use importance sampling to iteratively obtain an unbiased estimate of the optimal control sequence. Specifically, when sufficiently sampled, the optimal control policy derived from an initial distribution P and the optimal distribution P^* is equivalent. In practical applications, due to the limited number of samples, we can only approximate the expected value through numerical statistical methods. To achieve optimal control performance, an iterative updating approach can be employed to gradually align the control distribution with the optimal control distribution. For the i -th iteration, an iterative update law can be derived:

$$u_t^{i+1} = \frac{\sum_{k=1}^K [\omega(V^{(k)}; U^i, \Sigma^i) v_t^i]}{\sum_{k=1}^K [\omega(V^{(k)}; U^i, \Sigma^i)]},$$

Algorithm 1 MPPI Controller

Initialize:

θ : learned network parameters
 μ^0, σ^0 : initial parameters for $\mathcal{N}$
 N_{MPPI}, N_{RL} : number of sampled/policy trajectories
 J : MPPI iterations
 $h_\theta, d_\theta, R_\theta, Q_\theta, \pi_\theta$: world model

Input: s_t, H : current state, rollout horizon

```

1: Given the set of terminal states of the control process  $\mathbb{S}_{\text{end}}$ 
2: while  $s_t \notin \mathbb{S}_{\text{end}}$  do
3:   Encode state  $z_t \leftarrow h_\theta(s_t)$ 
4:   Initialize  $\mathcal{N}_0(\mu_0, (\sigma_0)^2 I)$ 
5:   for each iteration  $j = 1 \dots J$  do
6:     Sample  $N_{MPPI}$  control sequences of length  $H$  from
        $\mathcal{N}_{j-1}(\mu_{j-1}, (\sigma_{j-1})^2 I)$ 
7:     Sample  $N_{RL}$  control sequences of length  $H$  using
       policy  $\pi_\theta$  and  $d_\theta$ 
8:     Initial  $\phi_\Gamma = 0$ 
9:     for all  $N_{MPPI} + N_{RL}$  control sequences do
10:      for step =  $t, \dots, t + H - 1$  do
11:         $\phi_\Gamma = \phi_\Gamma + R_\theta(z_t, a_t)$ 
12:         $z_{t+1} \leftarrow d_\theta(z_t, a_t)$ 
13:      end for
14:       $\phi_\Gamma = \phi_\Gamma + Q_\theta(z_H, \pi_\theta(z_H))$ 
15:    end for
16:    Update  $\mathcal{N}_j(\mu_j, (\sigma_j)^2 I)$  using (18)
17:  end for
18:  Sample  $\{v_t^j, \dots, v_{t+H-1}^j\}$  from  $\mathcal{N}_j(\mu_j, (\sigma_j)^2 I)$ 
19:  Apply  $a_t = v_t^j$  as control input and obtain  $s_{t+1}, r_t$ 
20:   $\mathcal{B} \leftarrow \mathcal{B} \cup \{s_t, a_t, r_t, s_{t+1}\}$ 
21: end while

```

$$\begin{aligned} \text{diag}(\sigma_t^{i+1}) &= \sqrt{\frac{\sum_{k=1}^K [\omega(V^{(k)}; U^i, \Sigma^i)(v_t^i - u_t^i) \circ (v_t^i - u_t^i)]}{\sum_{k=1}^K [\omega(V^{(k)}; U^i, \Sigma^i)]}} \\ t &= t_0, \dots, t_0 + N - 1, \end{aligned} \quad (15)$$

Here, $\omega(V^{(k)}; U^i, \Sigma^i) = \phi_\Gamma^{(k)}$, and each $V^{(k)}$ is drawn from $\mathcal{N}(U^i, \Sigma^i)$, with K denoting the total number of sampled trajectories. The pseudo-code corresponding to these updates is provided in Algorithm 1, while Fig.4 offers a comprehensive illustration of the framework's structure and its interaction with the environment.

B. RL for Guided Sampling

To further improve the sampling efficiency of the MPPI controller and reduce its iteration count, we additionally trained a RL policy to guide MPPI sampling. It is a stochastic maximum entropy policy designed to optimize the objective function:

$$\mathcal{L}_p(\theta) = \mathbb{E}_{(s,a)_{0:H} \sim \mathcal{B}} \left[\sum_{t=0}^H \gamma^t \left(Q(z_t, \pi_\theta(z_t)) + \alpha \mathcal{H}(\pi_\theta(\cdot | z_t)) \right) \right], \quad (16)$$

where γ is the discount factor. $\mathcal{H}(\pi_\theta(\cdot | z_t))$, which is defined as $-\sum_{a \in \mathcal{A}} \pi_\theta(a | z_t) \log \pi_\theta(a | z_t)$, denotes the entropy of the distribution of the policy. The temperature α balances exploitation

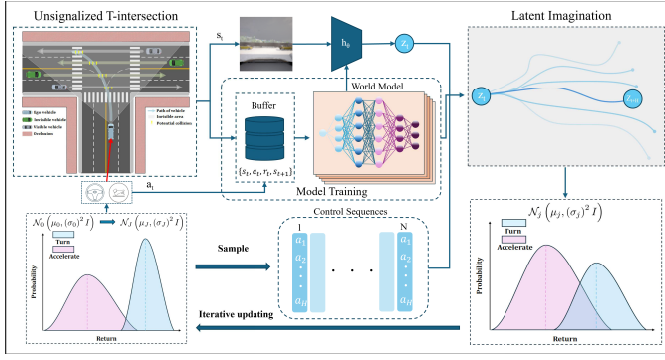


Fig. 4. The structure of the proposed framework.

of high-value actions and exploratory behavior. Unlike the SAC method, which integrates the entropy term into the value function updates, our approach focuses on maintaining the accuracy of Q-value estimates. During planning, we sample a batch of candidate sequences from π_θ alongside the Gaussian proposals in MPPI, merge them for trajectory rollouts, and use the results to update the MPPI distribution.

C. Risk-Aware Reward Formulation

We propose a risk-aware reward formulation for safe and effective autonomous driving in complex driving scenarios. Artificial potential fields (APFs) dynamically quantify interaction risks with traffic elements and are incorporated into a multi-objective reward function that evaluates candidate trajectories during planning to robustly guide the control policy.

1) *Risk Quantification*: To dynamically quantify driving risks, we design APFs that model traffic elements as field sources influencing surrounding space [29]. Notably, APFs are exclusively integrated into the reward function during training to guide policy convergence, not deployed during real-time control. The composite potential field E comprises three components:

$$E = w_1 E_V + w_2 E_L + w_3 E_B \quad (17)$$

- **Obstacle Vehicle Field (E_V)**: The obstacle vehicle potential field intensity depends on both static attributes (type, mass) and dynamic states (velocity, acceleration). Its model is defined as [30]:

$$\begin{aligned} E_V &= \sum_i^n M_i \lambda \frac{e^{-\beta a_i \cos \theta_i}}{|k_i|} \cdot \frac{k_i}{|k_i|} \\ &= \sum_i^n m_i \times (1.566 \times 10^{-14} v_i^{6.687} + 0.3345) \lambda \\ &\quad \times \frac{e^{-\beta a_i \cos \theta_i}}{\sqrt{[(x_i - x_0) \frac{\tau_x}{\rho v_i}]^2 + [(y_i - y_0) \tau_y]^2}} \cdot \frac{k_i}{|k_i|} \end{aligned}$$

Here, n is the number of obstacle vehicles. M_i is the equivalent mass of the obstacle vehicle, which depends on its velocity v_i and actual mass m_i . k_i is the pseudo-distance between the obstacle vehicle and the ego vehicle. (x_0, y_0) refers to the xy coordinates of the ego vehicle, while (x_i, y_i) is the xy coordinates of the obstacle vehicle. τ_x

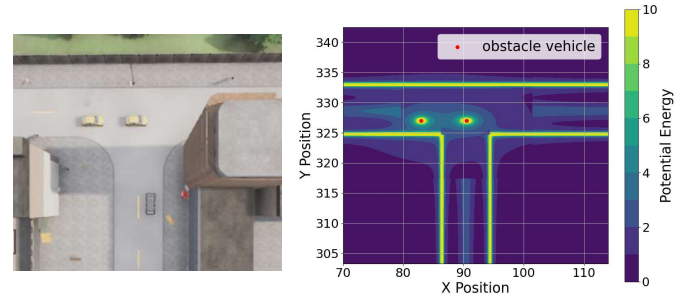


Fig. 5. Visualization of APFs at an unsignalized intersection. The red dots represent the centers of obstacle vehicles. The upper limit of the potential field energy is set to 10.

TABLE I
PARAMETERS OF ARTIFICIAL POTENTIAL FIELDS

Parameter	Value	Parameter	Value
λ	0.056	α	0.15
β	-0.169	A	8
τ_x	10	σ	0.8
τ_y	5	η	3

and τ_y are the safe thresholds for longitudinal and lateral safety distances, respectively. λ and β are undetermined coefficients, and θ is the clockwise angle between any point surrounding the obstacle vehicle and its center of mass with respect to the vehicle's direction of motion.

- **Lane Marking Field (E_L)**: Assuming lane lines and road boundaries are aligned with the x-axis, the potential field penalizes deviations from centerlines via Gaussian decay:

$$E_L = \sum_{i=1}^n A \cdot \exp \left(-\frac{(y_{L_i} - y_0)^2}{2\sigma^2} \right)$$

Here, A is the field intensity coefficient for each lane-marking type; n is the number of markings; y_{L_i} and y_0 denote the lateral positions of the i -th marking and the ego vehicle, respectively; and σ governs the spatial decay rate of the potential field.

- **Road Boundary Field (E_B)**: The boundary potential field model enforces drivable area containment through an inverse-square law:

$$E_B = \sum_{i=1}^n \eta \left(\frac{1}{(y_{B_i} - y_0)} \right)^2$$

APFs in Fig.5 can be constructed, and their parameters are shown in Table.I.

2) *Multi-Objective Reward Integration*: Building upon the risk foundations of the APFs, we formulate a comprehensive reward function at time t :

$$R_t = \underbrace{R_{APFs}}_{\text{Risk}} + \underbrace{R_{\text{prog}}}_{\text{Progress}} + \underbrace{R_{\text{lat}}}_{\text{Tracking}} + \underbrace{R_{\text{col}} + R_{\text{goal}}}_{\text{Event}}$$

- **Risk Potential Reward (R_{APFs})**: R_{APFs} represents the risk potential field reward, which quantifies the interaction risk between the ego vehicle and surrounding obstacles and is equal to $-E$ from (17).

- *Progress Reward* (R_{prog}): R_{prog} encourages the vehicle to move toward the goal by rewarding progress along the reference line. Let $\{p_i\}_{i=1}^N$ be the reference trajectory points, and let i_t be the index of the point p_i closest to the ego vehicle at t . Define

$$\Delta i_t = \max(i_t - i_{t-1}, 0),$$

then

$$R_{\text{adv}} = \min(\Delta i_t, r_{\text{cap}}).$$

Here, r_{cap} serves to cap the progress reward per time step, preventing the vehicle from exploiting the reward function by adopting excessively high speeds and thereby promoting safe driving behavior.

- *Lateral Penalty* (R_{lat}): R_{lat} penalizes deviations from the reference line to ensure stability. p_t is the position of the ego vehicle. Let

$$e_{\text{lat}} = \min_i \|p_t - p_i\|,$$

then

$$R_{\text{lat}} = -\min(e_{\text{lat}}, 1).$$

- *Event Rewards*: We employ two sparse, event-based rewards: a collision penalty R_{col} , which applies a negative value upon collision, and a goal reward R_{goal} , granted only upon successful task completion. They are defined as follows:

$$R_{\text{col}} = \begin{cases} -1, & \text{if a collision occurs,} \\ 0, & \text{otherwise,} \end{cases}$$

$$R_{\text{goal}} = \begin{cases} +1, & \text{if the goal is reached,} \\ 0, & \text{otherwise.} \end{cases}$$

IV. DRIVING-AWARE WORLD MODEL

Within our end-to-end driving framework, the Driving-Aware World Model (DAWM) serves as the perception and prediction backbone, tightly coupled with an RL-guided MPPI controller for efficient and robust control. DAWM's representation model embeds multi-scale spatial priors into a Vision Transformer (ViT) via a spatially-adaptive token reweighting scheme, sharpening focus on task-relevant regions. Its incremental latent dynamics model then predicts only the state increments—rather than full next states—enhancing stability and accuracy over long horizons. A principled uncertainty-aware loss weighting mechanism dynamically balances the multiple optimization objectives during DAWM training, accelerating convergence and improving overall performance. We next detail the design and implementation of each of these core modules.

A. Model Architecture

The DAWM consists of the following components [20]:

$$\begin{aligned} \text{Representation: } & z_t = h_\theta(s_t) \\ \text{Latent dynamics: } & z_{t+1} = d_\theta(z_t, a_t) \\ \text{Reward: } & \hat{r}_t = R_\theta(z_t, a_t) \end{aligned}$$

$$\text{Value: } \hat{q}_t = Q_\theta(z_t, a_t)$$

$$\text{Policy: } \hat{a}_t \sim \pi_\theta(z_t) \quad (18)$$

The representation model encodes the current observation into a compact latent state, which serves as the input for downstream modules. The latent dynamics model predicts state transitions in the latent space to support trajectory rollouts. The reward model estimates per-step rewards, while the value model approximates expected returns and provides learning signals for the policy model. The policy model generates action priors to initialize and steer MPPI sampling toward high-quality trajectories. These models work together to predict the dynamics of the environment. This modular design effectively enhances the representational capacity and task adaptability of the world model.

In terms of model structural design, all models except the representation model are implemented using Multi-Layer Perceptrons (MLPs). This approach not only meets the non-linear feature mapping needs of each model but also offers high computational efficiency and scalability.

B. Spatially-Adaptive Representation Model

Accurate modeling of dynamic interactions and spatial relationships among diverse traffic elements—such as vehicles and lane markings—remains a core challenge for representation in autonomous driving. Most world models employ convolutional representation models, which excel at extracting local spatial features but suffer from fixed receptive fields and therefore struggle to capture long-range dependencies (e.g., relative positions of distant vehicles or global road layout). ViT-based representation models leverage self-attention to overcome this limitation, yet their uniform patch partitioning and fixed positional encodings lack a strong spatial inductive bias, making it difficult to focus on task-relevant regions in driving scenarios.

We introduce an adaptive token reweighting scheme, guided by multi-scale spatial priors, to enhance standard ViT architectures with improved spatial adaptability. As illustrated in Fig. 6, our framework first processes input frames through an Efficient Multi-Scale Attention (EMA) module, which employs cross-space pooling and depthwise convolutions to generate pixel-level saliency maps highlighting task-relevant features (e.g., vehicle contours, lane markings). These maps are adaptively downsampled to the patch grid through pooling, producing spatial attention weights. During patch embedding, a ScoreNet computes per-token scores. The spatial weights and ScoreNet outputs are then combined to adjust patch embedding magnitudes, explicitly emphasizing critical regions while reducing background influence. The reweighted embeddings are processed by multi-head self-attention, enabling these spatially-enhanced features compensate for ViT's inherent lack of strong spatial inductive bias while preserving its capability to model long-range dependencies.

This spatially-adaptive reweighting mechanism complements ViT's self-attention by injecting multi-scale spatial priors and enhancing patch-level adaptability. Specifically, the EMA module contributes cross-scale saliency cues that enrich ViT's representational capacity with fine-grained detail and global context awareness. The adaptive reweighting

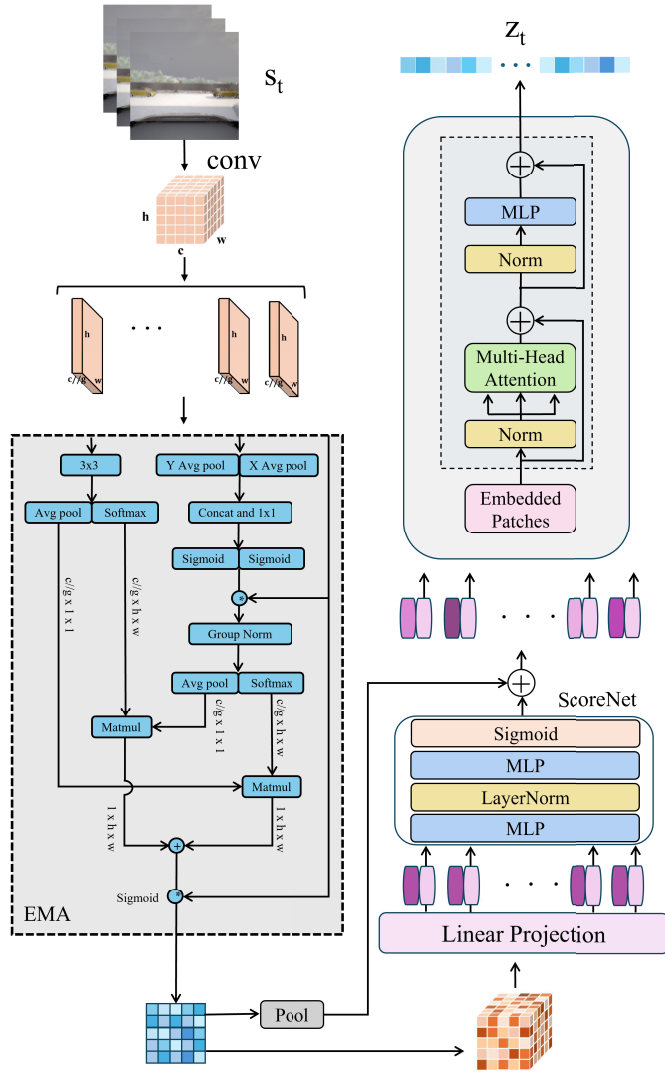


Fig. 6. The structure of the representation model. We first merge the input frames via a convolution and refine the result with EMA to obtain both enhanced features and a spatial prior. The refined feature map is split into patches, whose embeddings are reweighted by ScoreNet guided by the EMA prior, and then fed into a ViT to produce the latent state.

process—combining ScoreNet outputs with pooled spatial maps—provides a content-aware alternative to fixed positional encodings, allowing the representation model to dynamically emphasize task-relevant regions. Moreover, by suppressing less informative patches, our method reduces redundant attention computation and stabilizes training. Collectively, this integration improves ViT’s ability to perceive and represent key traffic elements under complex, dynamic conditions, leading to more accurate and robust downstream tasks.

C. Incremental Latent Dynamics Model

Traditional latent dynamics models directly map the current latent state to the next state. However, this “full-state” mapping approach faces two main issues: (i) since the model must learn the entire state transition, it is prone to overfitting or underfitting in complex dynamic environments; (ii) direct mapping struggles to capture subtle state changes, especially under

high-speed or highly dynamic maneuvers, leading to inaccurate predictions. Furthermore, as the representation model yields increasingly fine-grained and high-dimensional latent features, the dynamic model’s capacity requirements grow, exacerbating these shortcomings.

To address these issues, we introduce a residual connection into the dynamic model, reformulating the task from learning a full state mapping to predicting only the incremental change in the latent state. This residual design simplifies learning by focusing on smoother, small-magnitude transitions, thereby improving training stability and convergence speed. The residual pathway also enhances gradient flow and model robustness in unseen scenarios. Moreover, by decoupling global state updates from fine-grained increments, the model efficiently handles the increasingly complex features provided by the representation module, capturing subtle variations while suppressing noise. This mechanism not only improves predictive accuracy in complex environments but also enables tighter integration between representation and dynamics, yielding precise and robust predictions in complex driving scenarios.

D. Uncertainty-Aware Loss Weighting

We adopt a time-discounted, multi-task loss to jointly train reward prediction, value estimation, and latent-state consistency in an end-to-end fashion. Drawing on prior work [31], we weight each per-step loss by a discount factor λ^t and sum over horizon H :

$$\mathcal{L} = \mathbb{E}_{(s,a,r,s')_{0:H} \sim \mathcal{B}} \left[\sum_{t=0}^H \lambda^t (L_{c,t} + L_{r,t} + L_{v,t}) \right],$$

$$L_{c,t} = \|d_{\theta}(z_t, a_t) - \text{sg}(h(s'_t))\|_2^2, \quad L_{r,t} = \text{CE}(R_{\theta}(z_t, a_t), r_t), \\ L_{v,t} = \text{CE}(Q_{\theta}(z_t, a_t), q_t),$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator, $\text{CE}(\cdot, \cdot)$ is the cross-entropy loss, q_t the TD target, and z_t the latent state at step t . The consistency loss L_c ensures alignment between the representation model and the latent dynamics model, the reward loss L_r drives accurate reward prediction, and the value loss L_v fits the Q -value update against its TD target. This formulation transparently decomposes and balances all three objectives under a unified, time-discounted expectation.

Stable convergence in multi-task learning demands careful balancing of loss terms that differ widely in scale, optimization dynamics, and convergence speed. Prior methods addressed this by assigning fixed, manually tuned weights to each task—a practice that requires extensive hyperparameter sweeps, significantly inflating both training time and human effort—and once fixed, cannot accommodate evolving task difficulty, data noise fluctuations, or shifting priorities during training [20], [31]. To reduce manual tuning, Kendall et.al. introduced a homoscedastic uncertainty weighting scheme that scales each loss by the inverse of its estimated noise variance [23]. However, relying on a single variance estimate without online correction renders the approach vulnerable to initialization bias and subsequent overfitting [32].

To address these issues, we propose a novel uncertainty-aware weighting mechanism. Our approach rests on three

core principles: first, inspired by maximum likelihood estimation and Bayesian motivations for uncertainty modeling, we employ trainable log-scale parameters to dynamically capture homoscedastic uncertainty, thereby avoiding the limitations of fixed noise assumptions; second, leveraging the entropy-smoothing effect of temperature-controlled Softmax, we normalize these uncertainties into a balanced confidence distribution that flexibly trades off between focus and fairness across tasks; and third, we introduce a learnable global scaling factor with dedicated regularization applied separately to both uncertainty and scaling parameters. This design discourages both parameter collapse and divergence, ensuring all weights remain well-conditioned throughout training. We next elaborate on these components in detail.

Based on homoscedastic uncertainty, we first learn per-task log-scales α_i and map them to nonnegative noise estimates via

$$\sigma_i = \text{softplus}(\alpha_i),$$

Here, each α_i serves as a trainable proxy for the log-scale of the noise estimate σ_i , and the softplus mapping guarantees $\sigma_i > 0$ with smooth gradients [33]. These α_i are then converted into normalized confidences through a fixed-temperature Softmax,

$$p_i = \frac{\exp(-\alpha_i/T)}{\sum_{j=1}^3 \exp(-\alpha_j/T)},$$

which implicitly controls the sharpness of the weight distribution [34]. To modulate the overall gradient budget, we introduce a learnable scale β and set

$$s = \text{softplus}(\beta), \quad w_i = s \cdot p_i.$$

Finally, we assemble our total loss as

$$\mathcal{L} = \sum_{i=1}^3 w_i L_i + \lambda \left(\sum_{i=1}^3 \left(\sigma_i + \frac{1}{\sigma_i} \right) + \frac{1}{s^2} \right),$$

where a single regularization coefficient λ ties together the elastic penalties on both uncertainty and scale.

Overall, this uncertainty-aware weighting mechanism provides a systematic approach to stabilizing multi-task world model optimization. By adaptively reconciling task-relevant uncertainties and global gradient scaling, it reduces reliance on manual hyperparameter tuning while ensuring efficient and robust convergence. This design further strengthens the reliability of the proposed end-to-end framework, enabling consistent performance across diverse training scenarios.

V. EXPERIMENTAL RESULTS

A. Environment Setup

The experiments were conducted using the CARLA simulator, with the simulation scenario set at an unsignalized T-intersection in Town 1. CARLA leverages the NVIDIA PhysX Vehicle module within Unreal Engine 4 to simulate rigid-body dynamics, spring-damper suspensions, raycast-based tire forces, and aerodynamic drag. These physical components have been validated in prior studies for urban driving scenarios, demonstrating sufficient fidelity for controller development and evaluation [35], [36]. The simulation

ran in synchronous mode at a control frequency of 20 Hz. To simulate realistic vehicle interactions, two obstacle vehicles were introduced, forming a fleet randomly distributed along the lane within a 20-meter range, ensuring diversity in queue formation and interaction complexity. The departure times of these vehicles were uniformly sampled within the first two seconds of the simulation, increasing the likelihood of dynamic interactions with the ego vehicle at the intersection. To enhance realism, the obstacle vehicles' speeds were randomly sampled between 4m/s and 8m/s, with the following vehicle dynamically adjusting its speed based on the leading vehicle's motion, preventing collisions while preserving naturalistic driving behavior. This setup effectively replicates diverse and evolving interaction scenarios in dynamic traffic environments, enabling a robust evaluation of the proposed algorithm's adaptability and robustness.

The input to the model consisted of the vehicle's front-view images from the past three consecutive time frames. Each image had a dimension of (3, 128, 128), and the three images were concatenated along the time dimension, resulting in a combined input of dimension (9, 128, 128). The output consisted of vehicle control commands, including acceleration (range: $(-3m/s^2, 3m/s^2)$) and steering angle (range: $(-60^\circ, 60^\circ)$). The experimental hardware environment included an NVIDIA A40 GPU and an Intel(R) Xeon(R) Gold 5318Y CPU.

B. Baselines

1) *TD-MPC2*: TD-MPC2 is an improved version of TD-MPC, featuring a more refined model architecture and loss design that enhance training efficiency [20], [31]. As a representative model-based reinforcement learning algorithm, TD-MPC2 demonstrates higher stability and better control performance in complex dynamic environments, making it a strong benchmark for evaluating model-based methods.

2) *SAC*: Soft Actor-Critic (SAC) is a representative model-free reinforcement learning algorithm [37]. It maximizes policy entropy to encourage exploration, ensuring policy diversity and robustness. Due to its high data efficiency, robust performance, and ease of training, SAC serves as an essential baseline for evaluating model-free methods.

3) *MPPI-Sim*: MPPI-sim refers to the MPPI control method applied in an idealized simulation environment with access to ground-truth data [24]. This includes precise information about the ego vehicle's kinematics, as well as complete traffic element coordinates. APFs modeling in MPPI-sim leverages these known parameters. To ensure a fair comparison, the hyperparameters of MPPI-sim are set to match those of the proposed algorithm, with the hyperparameter categories (e.g., prediction horizon, sample count, noise covariance) detailed in Table II.

C. Model Architecture and Parameters

Our DAWM model comprises five components: a representation model, a latent dynamics model, a reward model, a value model, and a policy model. The representation model first applies a 5×5 convolution with stride 2 (output

TABLE II
MPPI HYPERPARAMETERS USED IN OUR EXPERIMENTS

Hyperparameter	Value
Iteration number J	6
Prediction horizon H	3
Inverse temperature λ	2
Number of RL rollouts N_{RL}	25
Number of MPPI rollouts N_{MPPI}	512

TABLE III
INPUT AND OUTPUT SHAPES OF EACH DAWM COMPONENT

Component	Input	Output
Representation model	$x_t \in \mathbb{R}^{9 \times 128 \times 128}$	$z_t \in \mathbb{R}^{512}$
Latent dynamics model	$[z_t; a_t] \in \mathbb{R}^{514}$	$z_{t+1} \in \mathbb{R}^{512}$
Reward model	$[z_t; a_t] \in \mathbb{R}^{514}$	logits $\in \mathbb{R}^{101} \rightarrow \mathbb{R}^1$
Value model	$[z_t; a_t] \in \mathbb{R}^{514}$	logits $\in \mathbb{R}^{101} \rightarrow \mathbb{R}^1$
Policy model	$z_t \in \mathbb{R}^{512}$	$[\mu, \log \sigma] \in \mathbb{R}^{2 \times 2}$

channels =32) to the stacked input frames $x_t \in \mathbb{R}^{9 \times 128 \times 128}$, then passes the result through a grouped EMA module (groups =4) to produce feature maps $F \in \mathbb{R}^{32 \times 64 \times 64}$ and spatial weight maps W . The feature maps F are converted into P patch tokens by a patch embedding layer (patch size =8, embedding dimension =256), yielding $T \in \mathbb{R}^{P \times 256}$. The ScoreNet—implemented as two linear layers with sigmoid activation—assigns each token in T a scalar importance score in $[0, 1]$. A single-layer Vision Transformer (attention heads =8, embedding dimension=512) processes the reweighted tokens to capture long-range dependencies before projecting to the latent state.

The dynamics, reward, value, and policy models share a common backbone of three fully connected hidden layers (512 units each, Mish activation). The dynamics model adds a skip connection from z_t to its final layer output, producing the next latent state $z_{t+1} \in \mathbb{R}^{512}$. Both the reward and value models append a 101-unit output layer whose logits are converted via softmax and an expectation over fixed support points into a scalar (following DreamerV3 [38]). The policy network outputs two 2-dimensional vectors, denoted as $\mu \in \mathbb{R}^2$ and $\log \sigma \in \mathbb{R}^2$, which serve as the mean and log-standard deviation parameters of the 2-dimensional Gaussian action distribution. Table.III summarizes the input and output shapes for each of the components described above.

TD-MPC2 shares the same architecture as DAWM, and all parameters, except those of the representation model, are identical to DAWM's. The representation model stacks four convolutional layers with configurations (32, 7, 2), (32, 5, 2), (32, 3, 2) and (32, 3, 1) respectively (channels, kernel size, stride). Feature maps are then reduced to a 4×4 grid by adaptive average pooling and flattened into a 512-dimensional vector for downstream processing. The critic and policy models in SAC adopt the same architecture as in the DAWM, featuring three fully connected hidden layers each with 512 units and using the Mish activation function.

D. Training Performance Comparison

The results of all experiments are averaged over three independent trials, with each trial consisting of 300,000



Fig. 7. The experimental environment was a signal-free intersection in CARLA. The ego vehicle is represented by the black car, while the two yellow cars serve as obstacle vehicles.

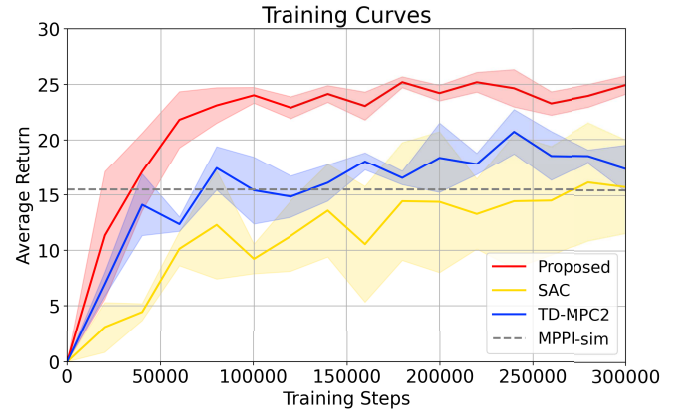


Fig. 8. The return curves of different algorithms. The solid lines represent the mean values over 3 evaluations, and the shaded areas indicate the standard deviation. Each evaluation consists of 30,000 steps, with the average return recorded every 20,000 steps.

environment steps. The evaluation curves shown in Fig.8 depict the training performance of TD-MPC2, SAC, MPPI-sim, and the proposed algorithm, in terms of average return over training steps. During the convergence phase, fluctuations are observed, likely due to the inherent randomness in the simulated environment, such as the stochastic initial positions of obstacles and departure times.

Despite these fluctuations, the proposed algorithm consistently outperforms the baseline methods, achieving faster convergence and higher average returns. These results highlight the superior sample efficiency and robustness of the proposed approach compared to TD-MPC2, SAC, and MPPI-sim.

E. Performance Evaluation

In this section, we evaluate the training results of different algorithms through multiple test episodes and analyze their performance across various driving scenarios. Specifically, the evaluation is conducted using the model checkpoints corresponding to the highest average return observed during training. To comprehensively assess performance, we

TABLE IV
EVALUATION OF PERFORMANCE METRICS

Algorithm	Success rate (%)	Collision rate (%)	risk value	Deviation (cm)	Computation Time (ms)
TD-MPC2	79.7 ± 2.1	5.0 ± 0.8	6.27 ± 0.05	12.4 ± 0.2	12.0 ± 0.2
SAC	67.3 ± 3.7	25.7 ± 1.9	7.66 ± 0.13	17.8 ± 0.2	1.4 ± 0.1
MPPI-sim	32.3 ± 1.2	67.7 ± 3.3	14.12 ± 0.04	3.1 ± 0.1	24.4 ± 0.2
Proposed	99.0 ± 0.8	0.7 ± 0.5	6.14 ± 0.03	7.7 ± 0.1	14.9 ± 0.1

randomly generate 100 unique driving scenarios, varying the initial positions and velocities of the obstacle vehicles.

Performance is assessed using five key metrics: success rate, collision rate, deviation from lane center, risk value, and computation time. Success rate measures the percentage of episodes in which the autonomous vehicle successfully reaches its destination without collisions. Collision rate reflects the percentage of episodes in which collisions occur. Deviation from lane center represents the average distance of the ego vehicle's trajectory from the center of the lane, where smaller deviations indicate more stable lane-keeping. The risk value is derived from the average risk potential field value received by the ego vehicle at each environment step, with smaller values indicating better safety margins. In addition, the computation time is measured as the average time required to generate a single control command at each step, reflecting the real-time capabilities and computational efficiency of the algorithms. Together, these metrics provide a holistic evaluation of the algorithm's performance in diverse driving conditions. The mean and standard deviation of the results from three repeated tests are summarized in Table IV.

From Table IV, it can be observed that although MPPI-sim has full knowledge of the ego vehicle's kinematics model and surrounding environment information, resulting in significantly smaller lane center deviation, its success rate is still below 50%. This indicates that despite MPPI-sim generating relatively accurate control actions, the MPPI method is prone to getting stuck in local optima in high-dynamic scenarios, leading to collisions with the obstacle vehicle. In contrast, our proposed method enhances environmental prediction capabilities leveraging DAWM while also guiding the MPPI optimizer with a reinforcement learning strategy. By approximating the terminal cost function with the learned value function, we extend the solution of MPPI from a local optimum to an approximate global optimum, which significantly reduces the collision rate and enhances the control performance of MPPI.

The evaluation results show that the proposed method achieves a significantly higher success rate compared to other baseline approaches, with the lowest collision rate among all evaluated algorithms. These findings demonstrate the superior performance and robustness of the proposed method in handling diverse driving scenarios. Furthermore, the method exhibits the smaller lateral deviation from the lane center, reflecting its ability to maintain accurate and stable control during navigation. Lastly, the smallest risk value highlights enhanced safety, as the algorithm ensures larger margins to avoid potential collisions. Although our method incurs a modest increase in computation time due to a more complex

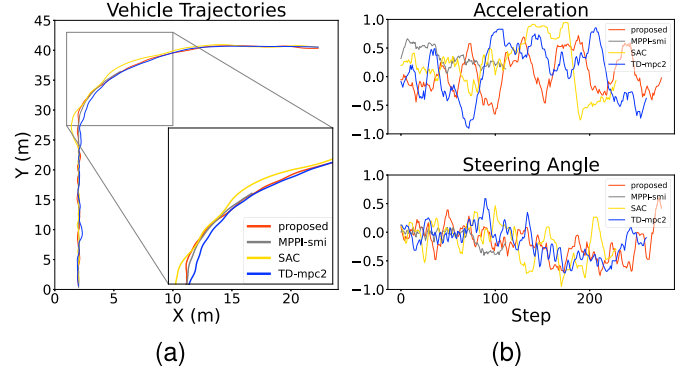


Fig. 9. (a) Vehicle trajectories for our method and baselines. (b) Corresponding normalized control inputs, smoothed using an 11-step sliding window.

network architecture, it still maintains real-time capability, reflecting a practical balance between computational cost and performance gains.

To visually demonstrate the effectiveness of the proposed method, Fig. 9 presents the trajectories and corresponding control signals generated by each algorithm under the same driving scenario. Fig. 9(a) shows vehicle trajectories under the same scenario. The MPPI baseline collides with an obstacle and terminates prematurely. Among the methods that complete the task, our approach exhibits the smallest deviation from the reference path during the straight-driving phase and the smoothest trajectory throughout the interactive phase. Fig. 9(b) depicts control inputs over time, where our method's acceleration and steering signals are smoother and more stable. These results indicate that the proposed algorithm achieves more stable and reliable control performance.

F. Ablation Study

Ablation studies are conducted to analyze the contribution of each component in the proposed DAWM and demonstrate their significance in improving overall performance. By systematically removing or modifying key components, we form several baseline models to compare against the full DAWM framework. The training curves for different methods are shown in Fig. 10. The test results of these baselines are summarized in Table V, with detailed evaluations for the following variants:

1) *DAWM Without Spatial Reweighting*: This variant removes the spatial reweighting scheme, resulting in a plain ViT backbone. As shown in Table V, the absence of spatial adaptability leads to degraded performance across multiple metrics: the success rate drops significantly from 99.0% to

TABLE V
ABLATION EXPERIMENT RESULTS FOR PROPOSED ALGORITHM

Algorithm	Success rate (%)	Collision rate (%)	risk value	Deviation (cm)	Computation Time (ms)
Proposed w/o spatial reweighting	89.7 ± 4.7	7.7 ± 2.5	6.25 ± 0.02	13.6 ± 0.3	13.5 ± 0.1
Proposed w/o spatial priors	84.7 ± 0.4	12.3 ± 1.7	6.29 ± 0.08	8.7 ± 0.1	14.2 ± 0.2
Proposed w/o incremental prediction	60.7 ± 0.9	35.3 ± 1.7	7.27 ± 0.06	12.3 ± 0.2	15.2 ± 0.2
Proposed w/o weight adjustment	71.7 ± 2.6	24.0 ± 2.2	6.67 ± 0.06	10.5 ± 0.2	15.0 ± 0.1
Proposed	99.0 ± 0.8	0.7 ± 0.5	6.14 ± 0.03	7.7 ± 0.1	14.9 ± 0.1

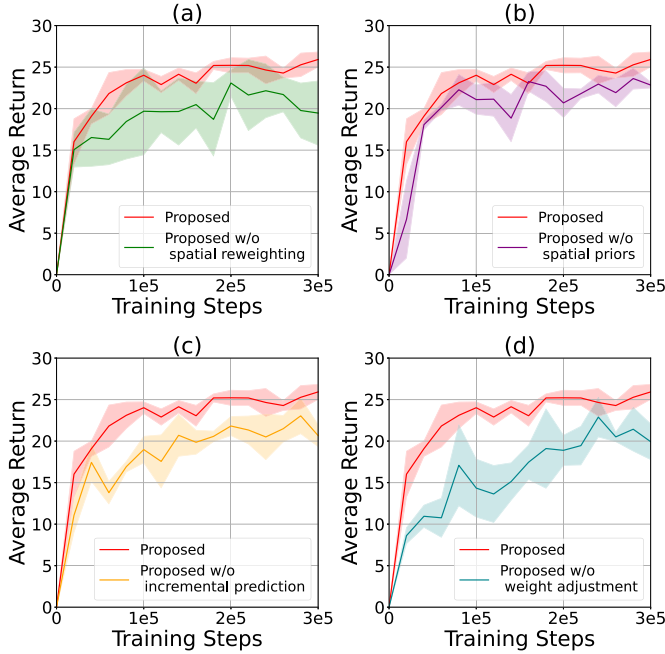


Fig. 10. Illustrates the training curves of the proposed algorithm compared with the four ablation variants. Subplots (a), (b), (c), and (d) correspond to the variants without spatial reweighting, without multi-scale spatial priors, without incremental prediction, and without dynamic weight adjustment, respectively.

89.7%, and the collision rate increases by over an order of magnitude (from 0.7% to 7.7%), accompanied by rises in risk values and lane deviation distance. These results confirm that adaptive token reweighting plays a key role in highlighting task-relevant regions while reducing background influence, thereby enhancing spatial adaptability in complex driving scenarios.

2) *DAWM Without Multi-Scale Spatial Priors*: This variant disables the multi-scale spatial priors derived from EMA and relies solely on the ScoreNet-generated weights without external guidance. While ScoreNet remains trainable, the absence of reliable spatial priors undermines its consistency and effectiveness, making it more prone to emphasizing unstable or irrelevant features. As a result, the overall model suffers from degraded downstream control performance, including a 14.3% drop in success rate and increased lane deviation and collision rates, as shown in Table V. These findings underscore the role of multi-scale spatial priors as crucial inductive biases that guide the reweighting process, enabling the model to form robust and interpretable spatial representations.

3) *DAWM Without Incremental Prediction*: This variant excludes the incremental latent dynamic model, which

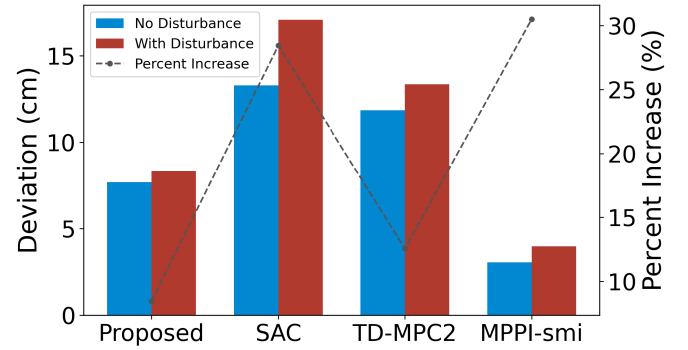


Fig. 11. Comparison of the mean lateral deviation and relative performance degradation of the proposed method and baseline algorithms under no disturbance and disturbance conditions.

was designed to iteratively refine predictions over time for improved temporal consistency and accuracy. The absence of this component leads to a 20% increase in lane deviation and a higher collision rate, as shown in Table V. These results demonstrate that incremental prediction enhances the model's ability to estimate latent states, which in turn enhances control stability and downstream training.

4) *DAWM Without Dynamic Weight Adjustment*: In this version, the uncertainty-aware weighting mechanism is replaced by fixed weights to evaluate the impact of adaptive weight learning. The training curves (Fig. 10(d)) show that dynamic weight adjustment not only improves convergence stability but also avoids local optima, resulting in a higher final success rate compared to fixed weights. This highlights the importance of balancing multiple objectives dynamically during training.

The ablation study results demonstrate that all components of DAWM (including the spatial reweighting scheme, EMA, the incremental latent dynamics model, and the uncertainty-aware weighting mechanism) significantly contribute to its superior performance. Each component enhances specific aspects of the driving policy, such as feature extraction, temporal consistency, and objective balancing, thereby validating the design choices underlying the proposed framework.

G. Control Stability

To evaluate the robustness of the RL-guided MPPI control strategy under action perturbations, we added zero-mean Gaussian noise with a standard deviation of 0.5 to the original action outputs, and then clipped the perturbed actions to the $[-1, 1]$ range. In 100 test scenarios, comparative experiments were conducted under both “No Disturbance” and “With Disturbance” conditions for our method (Proposed)

and three baseline algorithms (SAC, TD-MPC2, MPPI-sim). In each experiment, the mean lateral deviation of the vehicle was recorded, and the percentage increase in deviation under disturbance relative to the no-disturbance case was computed. Fig. 11 presents the performance comparison of each algorithm: bar charts depict the mean lateral deviation (in cm) under the two conditions, and the overlaid line plot shows the relative increase in deviation (in %). The results show that our proposed method achieves a mean deviation of only 7.68cm under no disturbance, which rises to 8.33cm when disturbance is applied—an increase of approximately 8.5%. In contrast, the deviation increases for SAC, TD-MPC2, and MPPI-SMI are 28.5%, 12.6%, and 30.5%, respectively. This comparison demonstrates that the MPPI controller, when coupled with the DAWM, maintains minimal performance degradation under action perturbations, exhibiting excellent stability and robustness.

VI. CONCLUSION AND FUTURE WORK

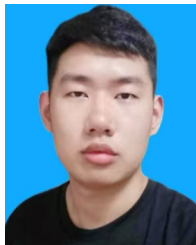
We present an end-to-end autonomous driving framework that integrates a Driving-Aware World Model (DAWM) with RL-guided MPPI. DAWM enhances spatial focus via adaptive token reweighting and improves prediction via incremental latent dynamics. A principled uncertainty-aware loss weighting further stabilizes multi-task training. Experiments in unsignalized T-intersections validate the framework's superior control performance, efficiency, and robustness.

In future work, we will integrate a simplified vehicle dynamics model into the data-driven world model and control pipeline under vehicle-constrained conditions to improve simulation fidelity and ease transfer to real vehicles. Concurrently, based on this simplified model and a small-perturbation assumption, we will perform a local Lyapunov analysis and, through linearization, construct approximate invariant sets, aiming to provide preliminary theoretical guarantees of local asymptotic stability or bounded convergence for the closed-loop system under the integrated MPPI and RL framework.

REFERENCES

- [1] W. Huang, H. Liu, Z. Huang, and C. Lv, "Safety-aware human-in-the-loop reinforcement learning with shared control for autonomous driving," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 11, pp. 16181–16192, Nov. 2024.
- [2] P. Wang, W. Hao, and Y. Jin, "Fine-grained traffic flow prediction of various vehicle types via fusion of multisource data and deep learning approaches," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 11, pp. 6921–6930, Nov. 2021.
- [3] X. Wang, K. Li, and A. Chehri, "Multi-sensor fusion technology for 3D object detection in autonomous driving: A review," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 2, pp. 1148–1165, Feb. 2024.
- [4] M. Al-Sharman, R. Dempster, M. A. Daoud, M. Nasr, D. Ray-side, and W. Melek, "Self-learned autonomous driving at unsignalized intersections: A hierarchical reinforced learning approach for feasible decision-making," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 11, pp. 12345–12356, Nov. 2023.
- [5] J. Luo et al., "Real-time cooperative vehicle coordination at unsignalized road intersections," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 5, pp. 5390–5405, May 2023.
- [6] C. Chen, Q. Xu, M. Cai, J. Wang, J. Wang, and K. Li, "Conflict-free cooperation method for connected and automated vehicles at unsignalized intersections: Graph-based modeling and optimality analysis," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 21897–21914, Nov. 2022.
- [7] S. Li, K. Peng, F. Hui, Z. Li, C. Wei, and W. Wang, "A decision-making approach for complex unsignalized intersection by deep reinforcement learning," *IEEE Trans. Veh. Technol.*, vol. 73, no. 11, pp. 16134–16147, Nov. 2024.
- [8] X. Lu, H. Zhao, C. Li, B. Gao, and H. Chen, "A game-theoretic approach on conflict resolution of autonomous vehicles at unsignalized intersections," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 11, pp. 12535–12548, Nov. 2023.
- [9] H. Shu, T. Liu, X. Mu, and D. Cao, "Driving tasks transfer using deep reinforcement learning for decision-making of autonomous vehicles in unsignalized intersection," *IEEE Trans. Veh. Technol.*, vol. 71, no. 1, pp. 41–52, Jan. 2022.
- [10] X. Tang et al., "Uncertainty-aware decision-making for autonomous driving at uncontrolled intersections," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 9, pp. 9725–9735, Sep. 2023.
- [11] H. Li, T. Zhang, S. Zheng, and C. Sun, "Distributed MPC for multi-vehicle cooperative control considering the surrounding vehicle personality," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 3, pp. 2814–2826, Mar. 2024.
- [12] M. Ammour, R. Orjuela, and M. Basset, "A MPC combined decision making and trajectory planning for autonomous vehicle collision avoidance," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 24805–24817, Dec. 2022.
- [13] M. S. Gandhi, B. Vlahov, J. Gibson, G. Williams, and E. A. Theodorou, "Robust model predictive path integral control: Analysis and performance guarantees," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 1423–1430, Apr. 2021.
- [14] P. C. N. Verheijen, V. Breschi, and M. Lazar, "Handbook of linear data-driven predictive control: Theory, implementation and design," *Annu. Rev. Control*, vol. 56, Jan. 2023, Art. no. 100914.
- [15] J. Liu et al., "Reinforcement learning-based high-speed path following control for autonomous vehicles," *IEEE Trans. Veh. Technol.*, vol. 73, no. 6, pp. 7603–7615, Jun. 2024.
- [16] X. Qi, L. Zhang, P. Wang, J. Yang, T. Zou, and W. Liu, "Learning-based MPC for autonomous motion planning at freeway off-ramp diverging," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 10, pp. 1–11, Oct. 2024.
- [17] Q. Li, X. Jia, S. Wang, and J. Yan, "Think2Drive: Efficient reinforcement learning by thinking in latent world model for quasi-realistic autonomous driving (in Carla-v2)," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 142–158.
- [18] Z. Gao et al., "Enhance sample efficiency and robustness of end-to-end urban autonomous driving via semantic masked world model," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 10, pp. 13067–13079, Oct. 2024.
- [19] D. Hafner et al., "Learning latent dynamics for planning from pixels," in *Proc. Int. Conf. Mach. Learn.*, K. Chaudhuri and R. Salakhutdinov, Eds., May 2019, pp. 2555–2565.
- [20] N. A. Hansen, H. Su, and X. Wang, "Temporal difference learning for model predictive control," in *Proc. Int. Conf. Machin. Learn.*, 2022, pp. 8387–8406.
- [21] C. Chen, Y.-F. Wu, J. Yoon, and S. Ahn, "TransDreamer: Reinforcement learning with transformer world models," 2022, *arXiv:2202.09481*.
- [22] Z. Li et al., "BEVFormer: Learning bird's-eye-view representation from LiDAR-camera via spatiotemporal transformers," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 3, pp. 2020–2036, Mar. 2025.
- [23] R. Cipolla, Y. Gal, and A. Kendall, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 7482–7491.
- [24] G. Williams, P. Drews, B. Goldfain, J. M. Rehg, and E. A. Theodorou, "Information-theoretic model predictive control: Theory and applications to autonomous driving," *IEEE Trans. Robot.*, vol. 34, no. 6, pp. 1603–1622, Dec. 2018.
- [25] E. Trevisan and J. Alonso-Mora, "Biased-MPPI: Informing sampling-based model predictive control by fusing ancillary controllers," *IEEE Robot. Autom. Lett.*, vol. 9, no. 6, pp. 5871–5878, Jun. 2024.
- [26] T. Kim, G. Park, K. Kwak, J. Bae, and W. Lee, "Smooth model predictive path integral control without smoothing," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 10406–10413, Oct. 2022.
- [27] Z. Zhu and H. Zhao, "A survey of deep RL and IL for autonomous driving policy learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 9, pp. 14043–14065, Sep. 2022.
- [28] R. S. Sutton, "Learning to predict by the methods of temporal differences," *Mach. Learn.*, vol. 3, no. 1, pp. 9–44, Aug. 1988.
- [29] Y. Zhang, B. Shuai, R. Zhang, C. Fan, and W. Huang, "Modeling and simulation of driving risk pulse field and its application in car following model," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 8, pp. 8984–9000, Aug. 2024.

- [30] L. Li, J. Gan, X. Ji, X. Qu, and B. Ran, "Dynamic driving risk potential field model under the connected and automated vehicles environment and its application in car-following modeling," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 1, pp. 122–141, Jan. 2022.
- [31] N. Hansen, H. Su, and X. Wang, "TD-MPC2: Scalable, robust world models for continuous control," 2023, *arXiv:2310.16828*.
- [32] L. Kirchdorfer, C. Elich, S. Kutsche, H. Stuckenschmidt, L. Schott, and J. M. Köhler, "Analytical uncertainty-based loss weighting in multi-task learning," in *Proc. DAGM German Conf. Pattern Recognit.*, 2025, pp. 344–361.
- [33] E. Ilg et al., "Uncertainty estimates and multi-hypotheses networks for optical flow," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 652–667.
- [34] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.
- [35] U. Schumann, R. Wynands, and S. Meyne, "Metrological validation of the UNREAL gaming engine for type testing of traffic monitoring devices," *Meas., Sensors*, vol. 38, May 2025, Art. no. 101466.
- [36] C. Gómez-Huélamo et al., "Train here, drive there: Simulating real-world use cases with fully-autonomous driving architecture in CARLA simulator," in *Proc. Workshop Phys. Agents*, 2020, pp. 44–59.
- [37] T. Haarnoja et al., "Soft actor-critic algorithms and applications," 2019, *arXiv:1812.05905*.
- [38] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering diverse domains through world models," 2023, *arXiv:2301.04104*.



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